Database Management System (DBMS)

Previous Year Question Paper – 2018
Time: 3 Hours Maximum Marks: 100

Note: This is a sample/dummy placeholder question paper generated for demonstration purposes only. It does not represent an official examination paper. Some questions repeat across years with different wording, as is common in real previous-year question compilations.

Section A – Short Answer Questions (Answer any 5)

Q1. What are the different states of a transaction? Explain with a neat diagram. *(5 marks each)*

Q2. Explain BCNF and give an example of a relation that is in 3NF but not in BCNF. *(5 marks each)*

Q3. What is the use of aggregate functions such as SUM, AVG, COUNT in SQL? *(5 marks each)*

Q4. What is the purpose of a trigger in SQL? Write a simple example. *(5 marks each)*

Q5. Explain the concept of serializability with respect to concurrent transactions. *(5 marks each)*

Section B – Medium Answer Questions (Answer any 4)

Q1. Explain the conditions for deadlock and describe how deadlock detection works. *(10 marks each)*

Q2. Discuss the advantages and limitations of using views in a database. *(10 marks each)*

Q3. How are cardinality ratios represented in an ER diagram? Explain with examples. *(10 marks each)*

Q4. Explain the difference between a stored procedure and a user-defined function. *(10 marks each)*

Section C – Long Answer Questions (Answer any 3)

Q1. What is data abstraction in DBMS? Explain its levels with examples. *(15 marks each)*

Q2. Discuss how a DBMS recovers from a system failure using transaction logs. *(15 marks each)*

Q3. How is an Entity-Relationship model translated into relational tables? Explain with an example. *(15 marks each)*

--- End of Question Paper ---